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Batch - B4

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CL & NLP
Assignment - 1

1) $x_t = 0.5$
$h_{t-1} = 0.2$
$c_{t-1} = 0.4$

LSTM eq's are →

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$$

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$$

$$\tilde{c}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c)$$

$$c_t = f_t \cdot c_{t-1} + i_t \cdot \tilde{c}_t$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o)$$

$$h_t = o_t \cdot \tanh(c_t)$$

a) forget gate:

$$z_f = (0.7)(0.5) + (0.3)(0.2) + 0.1$$
$$= 0.5100$$

$$f_t = \sigma(z_f)$$
$$= \frac{1}{1+e^{-0.5100}}$$
$$= 0.6248$$

b) Input gate :

$$z_i = (0.6)(0.5) + (0.4)(0.2) + 0$$
$$= 0.3900$$

$$i_t = \frac{1}{1 + e^{-0.3900}}$$
$$= 0.5939$$

c) Candidate cell state ($\tilde{C}_t$) :

$$z_c = (0.9)(0.5) + (0.2)(0.2)$$
$$= 0.4900$$

$$\tilde{C}_t = \tanh(0.4900)$$
$$= 0.4542$$

d) New cell state (C_t) :

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$$
$$= (0.6249)(0.4) + (0.5939)(0.4542)$$
$$= 0.5197$$

e) Output gate (o_t) :

$$z_o = (0.5)(0.5) + (0.5)(0.2) + 0.1$$
$$= 0.4500$$

$$o_t = \frac{1}{1 + e^{-0.4500}}$$
$$= 0.6106$$

Good Write

f) New hidden state (ht):

$h_t = o_t \cdot \tanh(c_t)$

$\tanh(0.5197) \approx 0.4775$

$h_t = (0.6166)(0.4775)$

$= 0.2915$

2) $D = 10$

$H = 32$

no. of gates = 4

a) Parameters $= 4(DH + H^2 + H)$

$P = 4((10)(32) + (32)(32) + 32)$

$= 5504$

b) $P = 4(DH + H^2 + 2H)$

$= 4(10 \cdot 32 + 32 \cdot 32 + 2 \cdot 32)$

$= 5632$

3) Scaling factor $= P^x$

a) $P = 0.9$

After 10 steps:

$P^{10} = (0.9)^{10}$

$= 0.3487$

After 50 steps:

$$f^{50} = (0.9)^{50}$$
$$= 0.0052$$

b) For $k = 50$,

When $f = 0.5$

$$f^{50} = (0.5)^{50}$$

$$f^{50} = 8.8818 \times 10^{-16}$$

When $f = 0.99$,

$$f^{50} = (0.99)^{50}$$
$$= 0.6050$$

c) When forget gate is 0.5, gradient becomes almost zero after 50 time steps, making it extremely difficult for the network to learn long-term dependencies. In contrast when the forget gate is 0.99, approx 60.50% of the gradient remains after 50 steps.

4) Given:

Duration = 3 years
days per year = 365
hours per day = 24
Input look back = 168 hours
Forecast horizon = 24 hours
features per hour = 3
Stride = 1 hour
Batch size = 64

a) Total no. of hourly records:

$N = 3 \times 365 \times 24$
$= 26280$

b) For stride = 1;

Windows $= N - L - H + 1$
$= 26280 - 168 - 24 + 1$
$= 26089$

c) for batch of 64 windows

$X: (64, 168, 3)$

Each target contains next 24 hourly load values.

$y: (64, 24)$

Good Write

d) $N_{train} = 0.70 \times 26290$

$= 18396$

$w_{train} = 18396 - 168 - 24 + 1$

$= 18205$

$$\text{Batches} = \left\lceil \frac{\text{Training window}}{\text{Batch size}} \right\rceil$$

$$= \left\lceil \frac{18205}{64} \right\rceil$$

$$= \lceil 284.453 \rceil$$

$= 285$ mini-batches per epoch

e) $P = 4(DH + H^2 + H)$

$= 4[(3)(64) + (64)^2 + 64]$

$= 17409$

5)

a) Min-Max normalization :

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

$$= \frac{95 - 20}{120 - 20}$$

$x' = 0.75$

b) Prediction = { Failure, $p \geq 0.5$; No Failure, $p < 0.5$

$0.83 \geq 0.5$

∴ Failure

c) TP = 30

FP = Predicted failures − TP

= 45 − 30

FP = 15

FN = Actual Failures − TP

= 40 − 30

FN = 10

Actual non failures = 1000 − 40

= 960

TN = Actual non failure − FP

= 960 − 15

= 945

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{30}{30+15} = \frac{30}{45} = 66.67\%$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{30}{30+10} = \frac{30}{40} = 75\%$$

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$$\text{F1 score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$= \frac{2 \times \frac{30}{45} \times \frac{30}{40}}{\frac{30}{45} + \frac{30}{45}}$$

$$= 70.59\%$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{30 + 945}{1000}$$

$$= 97.50\%$$

a)) It should generally prioritise recall, because failing to detect a genuinely failing pump can cause equipment damage, unexpected downtime or safety risks.

-) Accuracy can be misleading because ~~the~~ only 40 out of 1,000 cycles actually lead to failure, a model can be correct on most normal cycles but still miss important failures, therefore Recall should usually be prioritised because missing a real failure can cause equipment damage, downtime or safety risks.